

2 - Ansible, Terraform

Name of report: `Ansible_SCM_LAB_2_Nikita_Niakhai`

Course: DevOps

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Date submission:

The goal is to easily, reliably and quickly maintain different kinds of systems at once.

Other software configuration management (SCM) tools:

Ansible

SaltStack

Puppet

Chef

▼ Task 1 - Prerequisites

Prepared 3 (VMs) on my Windows Host Machine:

- Controller VM Ubuntu Desktop 22.04 with Ansible (aka "`ansible-controller`")
- Ubuntu Server (aka "`ubuntu prod`")
- Fedora (aka "`fedora dev`"), that is Red-Hat based system that uses different tools instead of Ubuntu, e.g. `sudo dnf upgrade --refresh`

Added network setting inside Virtual Box for new Host Only adapter #2 :

My adapter is 192.168.30.1/24

- IPv4 Address: 192.168.30.1
- IPv4 Network Mask: 255.255.255.0
- DHCP Server Address: 192.168.30.100
- Lower/Upper Address Bounds: 192.168.30.101 to 192.168.30.254

Added NAT and second adapter Host Only adapter #2 on all VMs:

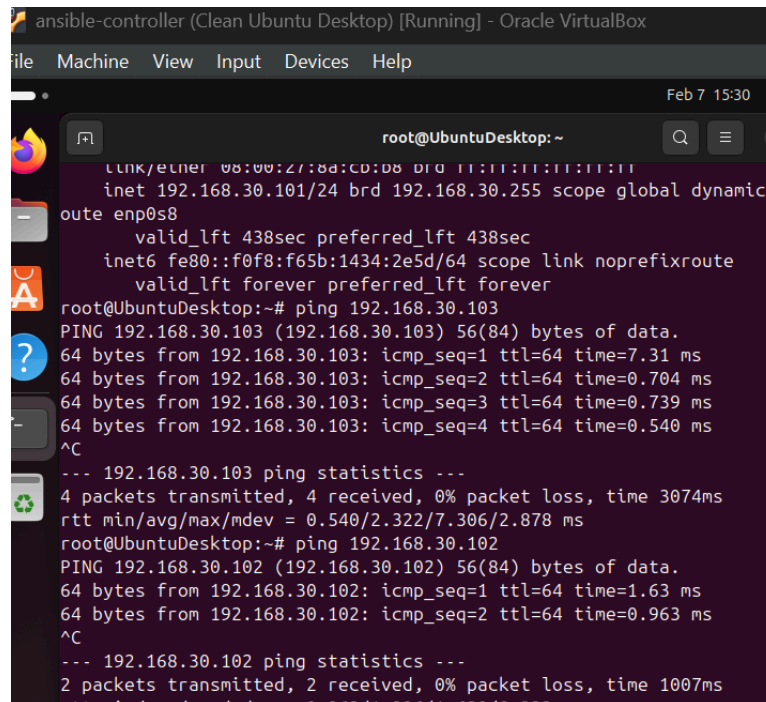
192.168.30.101 — `ansible-controller`

192.168.30.102 — `fedora dev`

192.168.30.103 — `ubuntu prod`

username to sign in: `ansible_user`

password: `user`



```
ansible-controller (Clean Ubuntu Desktop) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 7 15:30
root@UbuntuDesktop: ~
link/ether 08:00:27:8d:cd:08 brd ff:ff:ff:ff:ff:ff
inet 192.168.30.101/24 brd 192.168.30.255 scope global dynamic
out6 enp0s8
    valid_lft 438sec preferred_lft 438sec
inet6 fe80::f0f8:f65b:1434:2e5d/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
root@UbuntuDesktop:~# ping 192.168.30.103
PING 192.168.30.103 (192.168.30.103) 56(84) bytes of data.
64 bytes from 192.168.30.103: icmp_seq=1 ttl=64 time=7.31 ms
64 bytes from 192.168.30.103: icmp_seq=2 ttl=64 time=0.704 ms
64 bytes from 192.168.30.103: icmp_seq=3 ttl=64 time=0.739 ms
64 bytes from 192.168.30.103: icmp_seq=4 ttl=64 time=0.540 ms
^C
--- 192.168.30.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3074ms
rtt min/avg/max/mdev = 0.540/2.322/7.306/2.878 ms
root@UbuntuDesktop:~# ping 192.168.30.102
PING 192.168.30.102 (192.168.30.102) 56(84) bytes of data.
64 bytes from 192.168.30.102: icmp_seq=1 ttl=64 time=1.63 ms
64 bytes from 192.168.30.102: icmp_seq=2 ttl=64 time=0.963 ms
^C
--- 192.168.30.102 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1007ms
rtt min/avg/max/mdev = 0.963/1.296/1.630/0.333 ms
```

Figure. ansible machine showing his ip and then pinging dev and prod

Checked if python is installed on managable machines — true. Means I can run code from Ansible on nodes.

Then I opened ssh connections on managable nodes:

- Nodes:

- Open SSH server: Ubuntu `sudo apt install openssh-server -y`; Fedora `sudo dnf install openssh-server -y && sudo systemctl enable --now sshd`

- Controller:

- generated key `ssh-keygen`

```

/root/.ssh/authorized_keys
root@UbuntuDesktop:~# ls
key_ansible_lab  key_ansible_lab.pub  snap  vboxpostinstall.sh
root@UbuntuDesktop:~#

```

- Copied keys to nodes from controller

- empty key was not suitable for setting up this connection, so I created key on Fedora, because by default it is empty.
- `-i` to point to key file that should be sent.

```

root@UbuntuDesktop:~# ssh-copy-id -i /root/.ssh/authorized_keys/key_ansible_lab.pub liveuser@192.168.30.102

```

```

root@UbuntuDesktop:~/.ssh# ssh-copy-id -i ansible_ssh_master.pub nikita@192.168.30.103
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "ansible_ssh_master.pub"
The authenticity of host '192.168.30.103 (192.168.30.103)' can't be established.
ED25519 key fingerprint is SHA256:7jac37a2JudqwSvt1SXwLoETrkhtZEl0tHLZtdP05FM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out
any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted n
ow it is to install the new keys
nikita@192.168.30.103's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'nikita@192.168.30.103'"
and check to make sure that only the key(s) you wanted were added.
root@UbuntuDesktop:~/.ssh# ssh nikita@192.168.30.103

```

- Accessed ssh

```
ansible-controller (Clean Ubuntu Desktop) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 7 16:01

root@UbuntuDesktop: ~/.ssh
root@UbuntuDesktop:~/.ssh# whoami
root
root@UbuntuDesktop:~/.ssh# ssh liveuser@192.168.30.102
liveuser@192.168.30.102's password:
Activate the web console with: systemctl enable --now cockpit.socket

liveuser@localhost-live:~$ whoami
liveuser
```

```
ansible-controller (Clean Ubuntu Desktop) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Feb 7 16:05

nikita@LiveServerUbuntu24: ~
*** System restart required ***
nikita@LiveServerUbuntu24:~$ whoami
nikita
nikita@LiveServerUbuntu24:~$
nikita@LiveServerUbuntu24:~$
```

▼ Task 2 - SCM Theory

1. Ansible directory

- `ansible.cfg` is ini type file [\[ref.\]](#) that configures the ansible: sets limits, accesses, changes usage of SSH and etc.
- `inventory` directory [\[ref.\]](#) — is a place to define all managed nodes/hosts that controller will work with. It consists of groups for hosts, host names and their address (IP/domain name).

Inventories are of 2 types dynamic and static.

- Groups are created to apply the same playbooks on whole group of nodes, also to categorize nodes or gather them based on some fact, e.g. group of databases, or web-servers.
- `[group:vars]` — defines attributes that we need for each hosts, e.g. to connect to them (ssh keys location / password / username and etc.)
- `roles` directory [\[ref.\]](#)
 - Each role has a defined directory structure, all directories inside `/roles` folder represent each role. Inside each role you have following structure
 - `tasks` : tasks/job that are executed as a playbook
 - `defaults/group_vars` : - place for secrets.yaml with Vault secured / other shared values
 - `handlers` : kind of a place for common functions, e.g. you write there restart enable for some services, instead of creating separate tasks as steps in playbook
 - `templates` : some template files used to run anything for example html files, scripts and etc.
 - `files` : scripts, constable files, tables
 - `vars` : place for overall vars (used in a role)
- `meta` :
- `playbooks` folder
 - `-` are used to define items in the list, e.g. list of hosts, list of tasks, list of playbooks inside one file
 - `gather_facts` — allow ansible to gather information from the nodes, by default is `yes`
 - **colours** of executed commands:
 - `green` — nothing was changed
 - `red` — error

- **yellow** — something changes
- **blue** — command was skipped
- **hosts:** has name of manageable nodes + type of connection to this specific node:
 - Ansible has different types of connections to different types of nodes:
 - Network Devices — **network_cli** (type of ssh connection)
 - Linux Machines — **ssh**
 - Windows machines — **winrm** (is a protocol for establish secure remote connection, invented by Microsoft, based on WS-Management protocol; uses SOAP services, providing HTTP/S , yml)
 - Cloud machines — **local**
 - tasks have pre defined modules, each module requires specific params
 - there are keywords for each tasks, to be more precise in somethings, e.g.:
 - when — gets boolean value, e.g. from state received from `ansible_facts`
 - become: true — be root when you execute
- 2. The most important parameters from **ansible.cfg** :
- 3. The difference between `roles/.../vars` , `roles/.../defaults` and `playbook/.../group_vars` variables definitions.
- 4. Precedence order is specified in the docs. "Last defined wins"
 - Configuration settings
 - Command-line options
 - Playbook keywords
 - Variables
 - Direct Assignment

Ansible will make a change until it has to make a change! **Ansible is idempotent.**

▼ Task 3 - check github

Light tasks: Ansible Vault (definitely a good choice), Restart services

Hard: Certificates generation

Structure:

```
ansible-lab/
├─ ansible.cfg
├─ inventory/
│   └─ hosts.ini
├─ group_vars/
│   └─ all.yml           # plain vars
│   └─ secrets.yml       # Vault-encrypt
├─ roles/
│   └─ apache/
│       ├── defaults/
│       │   └─ main.yml
│       ├── handlers/
│       │   └─ main.yml
│       ├── tasks/
│       │   └─ main.yml
│       ├── templates/
│       │   ├── index.html.j2
│       │   └─ ports.conf.j2    # example custom config
│       └─ vars/
│           └─ main.yml         # if needed, but we'll u
se defaults mostly
└─ playbooks/
    └─ deploy_apache.yml
```

```

root@UbuntuDesktop:/home/nikita/Documents/ansible_lab/ansible-lab# cat inventory/hosts.ini
[dev]
192.168.30.102 ansible_host=192.168.30.102 ansible_pkg_mgr=dnf

[prod]
192.168.30.103 ansible_host=192.168.30.103 ansible_pkg_mgr=apt

[dev:vars]
env=dev

[prod:vars]
env=prod

[webservers:children]
dev
prod

```

Inventory file — hosts.ini showing both hosts (dev/prod)

```

root@UbuntuDesktop:/home/nikita/Documents/ansible_lab/ansible-lab# ansible webservers -m ping --ask-pass --ask-vault-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:
Vault password:
192.168.30.103 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Module invocation had junk after the JSON data:
192.168.30.102 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}

```

Test connectivity

```

ansible webservers -m ping --ask-vault-pass --ask-become-pass --ask-vault-pass

```



```

root@UbuntuDesktop:/home/nikita/Documents/ansible_lib/ansible-lab# ansible-playbook playbooks/deploy_certbot.yml
--ask-pass --ask-vault-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:
Vault password:

PLAY [Deploy Let's Encrypt certificates] *****

TASK [Gathering Facts] *****
ok: [192.168.30.103]
[WARNING]: Module invocation had junk after the JSON data:
ok: [192.168.30.102]

TASK [certbot : Install certbot (RedHat)] *****
skipping: [192.168.30.103]
[WARNING]: Module invocation had junk after the JSON data:
changed: [192.168.30.102]

TASK [certbot : Install certbot (Debian)] *****
skipping: [192.168.30.102]
ok: [192.168.30.103]

TASK [certbot : Open firewall port 80 (firewalld)] *****
skipping: [192.168.30.103]
[WARNING]: Module invocation had junk after the JSON data:
ok: [192.168.30.102]

TASK [certbot : Request Let's Encrypt certificate] *****
fatal: [192.168.30.103]: FAILED! => ({"changed": true, "cmd": ["certbot", "certonly", "--standalone", "--non-inter

```

```

...ignoring
[WARNING]: Module invocation had junk after the JSON data:
fatal: [192.168.30.102]: FAILED! => ({"changed": true, "cmd": ["certbot", "certonly", "--standalone", "--non-interactive", "--agree-tos", "--register-unsafely-without-email", "-d", "localhost-live.lab.local", "--staging"], "delta": "0:00:01.208816", "end": "2020-02-08 19:06:07.955892", "msg": "non-zero return code", "rc": 1, "start": "2020-02-08 19:06:06.747076", "stderr": "Saving debug log to /var/log/letsencrypt/letsencrypt.log\nAn unexpected error occurred:\nInvalid identifiers requested :: Cannot issue for 'localhost-live.lab.local': Domain name does not end with a valid public suffix (TLD)\nAsk for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letsencrypt.log or re-run Certbot with -v for more details.", "stdout_lines": ["Saving debug log to /var/log/letsencrypt/letsencrypt.log", "An unexpected error occurred:", "Invalid identifiers requested :: Cannot issue for 'localhost-live.lab.local': Domain name does not end with a valid public suffix (TLD)", "Ask for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letsencrypt.log or re-run Certbot with -v for more details."], "stdout": "Requesting a certificate for localhost-live.lab.local", "stdout_lines": ["Requesting a certificate for localhost-live.lab.local"]})
...ignoring

TASK [certbot : Show certbot result] *****
ok: [192.168.30.102] => {
  "msg": "Certbot failed (expected on private lab): Ask for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letsencrypt.log or re-run Certbot with -v for more details."
}
ok: [192.168.30.103] => {
  "msg": "Certbot failed (expected on private lab): Ask for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letsencrypt.log or re-run Certbot with -v for more details."
}

TASK [certbot : Set up certbot auto-renewal cron] *****
changed: [192.168.30.103]
[WARNING]: Module invocation had junk after the JSON data:
changed: [192.168.30.102]

TASK [certbot : Check certificate file] *****
ok: [192.168.30.103]
[WARNING]: Module invocation had junk after the JSON data:
ok: [192.168.30.102]

TASK [certbot : Report certificate status] *****
ok: [192.168.30.102] => {
  "msg": "Certificate not found (expected on private lab without public DNS)"
}
ok: [192.168.30.103] => {
  "msg": "Certificate not found (expected on private lab without public DNS)"
}

```

Run the playbook (the full ansible-playbook playbooks/deploy_certbot.yml)

```

ansible-playbook playbooks/deploy_apache.yml --ask-vault-pass

```

```

root@UbuntuDesktop:/home/nikita/Documents/ansible_lib/ansible-lab# ssh ansible_user@192.168.30.102
ansible_user@192.168.30.102's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Sun Feb  8 19:06:12 2026 from 192.168.30.101
ansible_user@localhost-live:~$ certbot --version
certbot 4.1.1
ansible_user@localhost-live:~$ exit
logout
Connection to 192.168.30.102 closed.
root@UbuntuDesktop:/home/nikita/Documents/ansible_lib/ansible-lab# ssh ansible_user@192.168.30.103
ansible_user@192.168.30.103's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-94-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Feb  8 07:22:29 PM UTC 2026

System load:          0.0
Usage of /:           16.2% of 24.44GB
Memory usage:         5%
Swap usage:           0%
Processes:            127
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe23:3529

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

28 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Feb  8 19:09:40 2026 from 192.168.30.101
ansible_user@LiveServerUbuntu24:~$ certbot --version
certbot 2.9.0

```

```

TASK [certbot : Report certificate status] *****
ok: [192.168.30.102] => {
  "msg": "Certificate not found (expected on private lab without public DNS)"
}
ok: [192.168.30.103] => {
  "msg": "Certificate not found (expected on private lab without public DNS)"
}

TASK [certbot : Force renew certificate] *****
changed: [192.168.30.103]
[WARNING]: Module invocation had junk after the JSON data:
changed: [192.168.30.102]

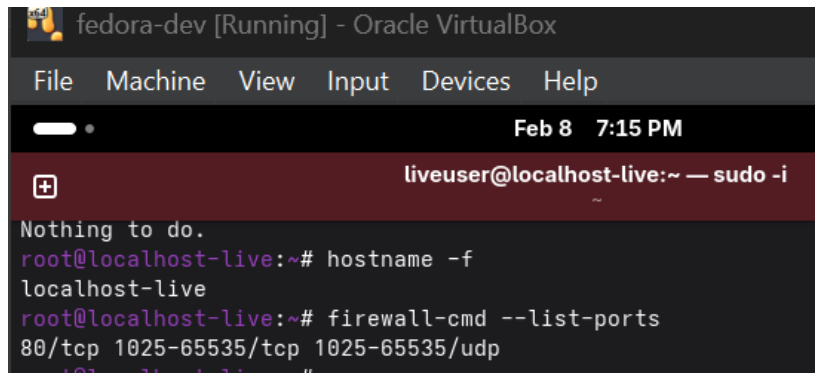
RUNNING HANDLER [certbot : Restart certbot-dependent services] *****
ok: [192.168.30.103] => {
  "msg": "Handler fired - no consuming service configured yet. Reload your web server / proxy here."
}
ok: [192.168.30.102] => {
  "msg": "Handler fired - no consuming service configured yet. Reload your web server / proxy here."
}

TASK [Show certificate path] *****
ok: [192.168.30.102] => {
  "msg": "Certificate at /etc/letsencrypt/live/localhost-live.lab.local/fullchain.pem"
}
ok: [192.168.30.103] => {
  "msg": "Certificate at /etc/letsencrypt/live/LiveServerUbuntu24.lab.local/fullchain.pem"
}

PLAY RECAP *****
192.168.30.102      : ok=11  changed=4  unreachable=0  failed=0  skipped=1  rescued=0  ignored=1
192.168.30.103      : ok=10   changed=3  unreachable=0  failed=0  skipped=2  rescued=0  ignored=1

```

Certbot installed — SSH into each host and run certbot --version

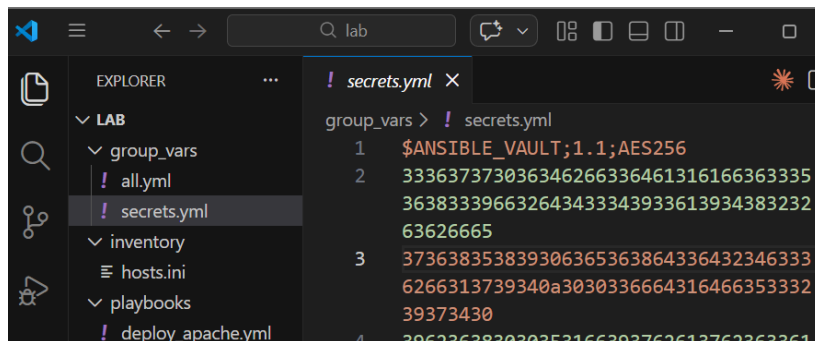


```
fedora-dev [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Feb 8 7:15 PM
liveuser@localhost-live:~ — sudo -i
Nothing to do.
root@localhost-live:~# hostname -f
localhost-live
root@localhost-live:~# firewall-cmd --list-ports
80/tcp 1025-65535/tcp 1025-65535/udp
root@localhost-live:~#
```

Firewall port open (Fedora only) — firewall-cmd --list-ports on 192.168.30.102

I encrypt the secrets file with Ansible Vault

The file `group_vars/secrets.yml` contains sensitive values in plain text right now. So I encrypt it before running the playbook.



```
EXPLORER
LAB
  group_vars
    ! secrets.yml
  inventory
    hosts.ini
  playbooks
    ! deploy_apache.yml
secrets.yml
group_vars > ! secrets.yml
1 $ANSIBLE_VAULT;1.1;AES256
2 333637373036346266336461316166363335
3 363833396632643433343933613934383232
4 63626665
5 373638353839306365363864336432346333
6 6266313739340a3030336664316466353332
7 39373430
8 396236383030353166393762613762363361
```

Vault file is encrypted — encrypted blob (not the plaintext)

Useful commands

Action	Command
Run playbook	<code>`ansible-playbook playbooks/deploy_apache.yml --ask-vault-pass`</code>

Run only on dev	`ansible-playbook play
books/deploy_apache.yml --ask-vault-pass -l dev`	
Run only on prod	`ansible-playbook play
books/deploy_apache.yml --ask-vault-pass -l prod`	
Restart Apache only	`ansible-playbook play
books/deploy_apache.yml --ask-vault-pass --tags restart`	
Edit vault secrets	`ansible-vault edit gr
oup_vars/secrets.yml`	
View vault secrets	`ansible-vault view gr
oup_vars/secrets.yml`	
Change vault password	`ansible-vault rekey g
roup_vars/secrets.yml`	
Check syntax	`ansible-playbook play
books/deploy_apache.yml --syntax-check`	
Dry run	`ansible-playbook play
books/deploy_apache.yml --ask-vault-pass --check`	
Test connectivity	`ansible webserver -m
ping --ask-vault-pass`	

▼ Bonus 2 - Ansible AWX

Comparison of open-source configuration management software

References

Ansible explained:

1. Indian guy - just the definition: www.youtube.com/watch?v=h8MurJBJVNc
2. Playbooks, but not all key semantics, but still good:
<https://www.youtube.com/watch?v=p9bda0-TIRc>